Opinion Formation on Networks

Alexander Pan

Mentors: Heather Z. Brooks, Franca Hoffmann, Mason A. Porter

Why do we care?

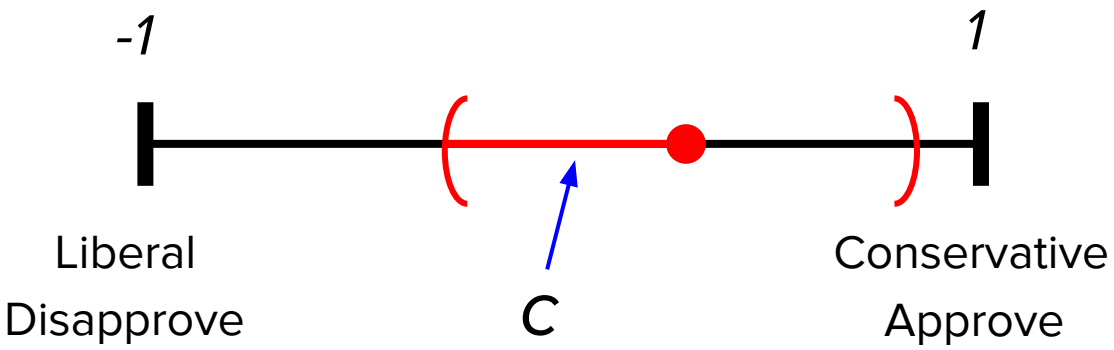
Media outlets and social media have a widespread impact on public opinion.

- Facebook --- Cambridge Analytica and Brexit
- Twitter --- Political bots in the 2016 presidential election
- Youtube --- Brazil and right wing extremism

How does media impact opinion in society?

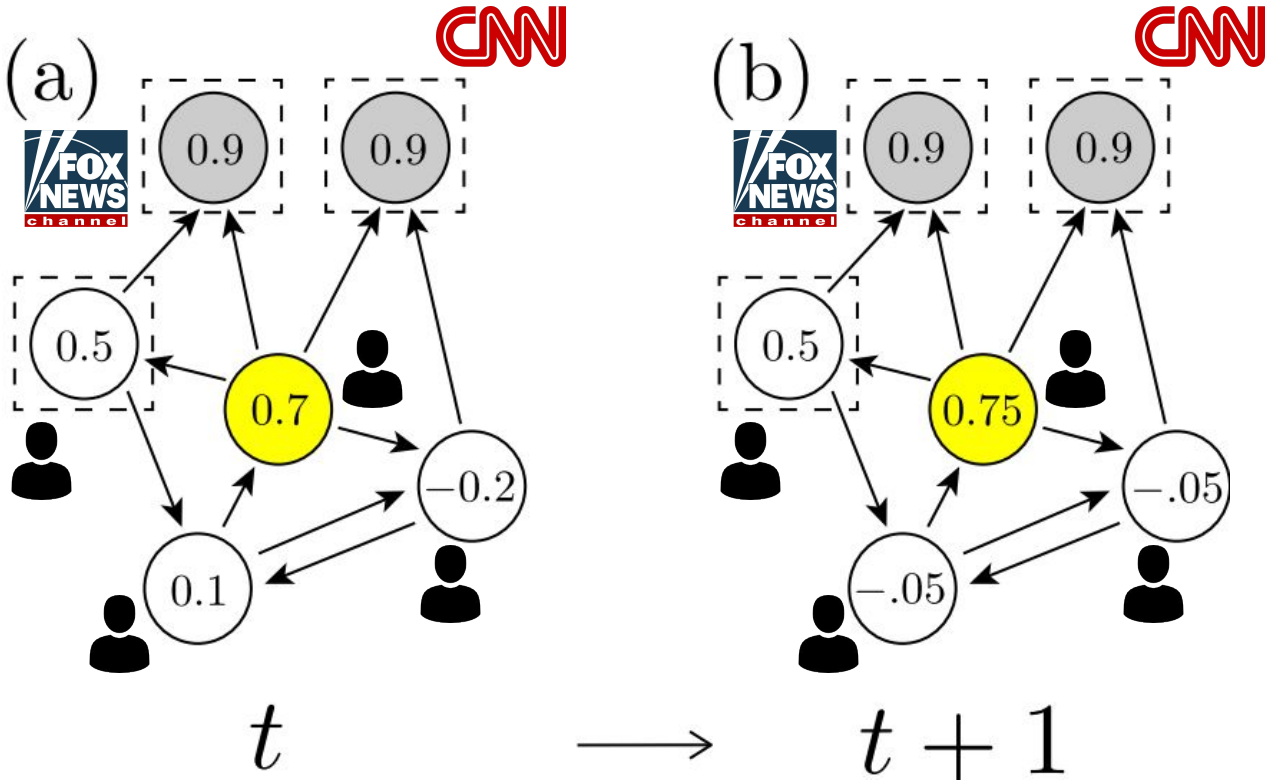
Concepts

- Interactions use **bounded-confidence opinion model**
- Population represented by **network**



[Brooks, Porter (preprint 2019)]

$$0.75 = \frac{1}{4}(0.7 + 0.5 + 0.9 + 0.9)$$

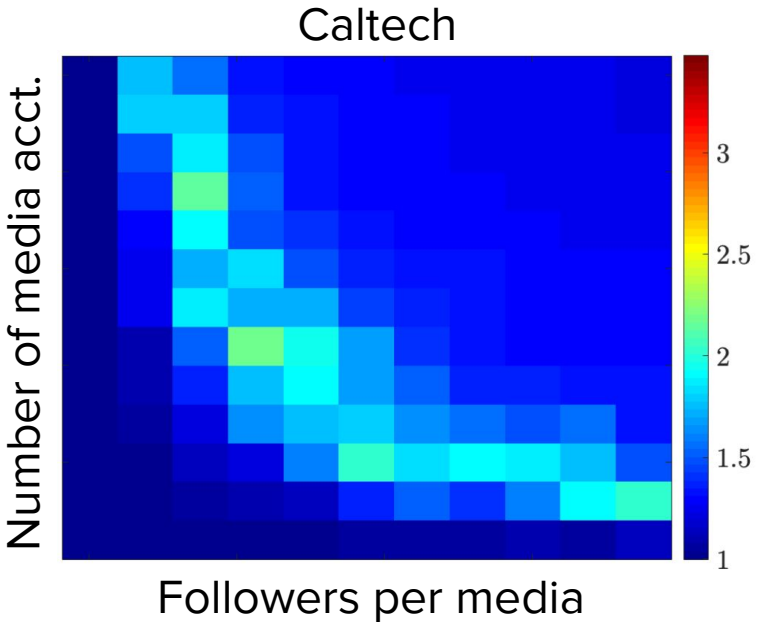


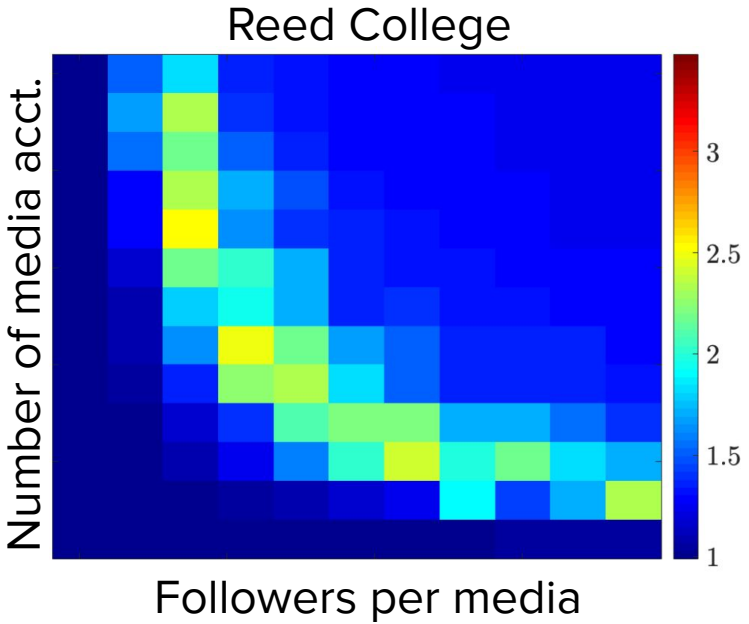
Nodes = individual or media, Edges = followership, Confidence bound c = 0.5

[Brooks, Porter (preprint 2019)]

Measuring media impact with *R*

R = (Avg. distance w/o media) / (Avg. distance w/ media)

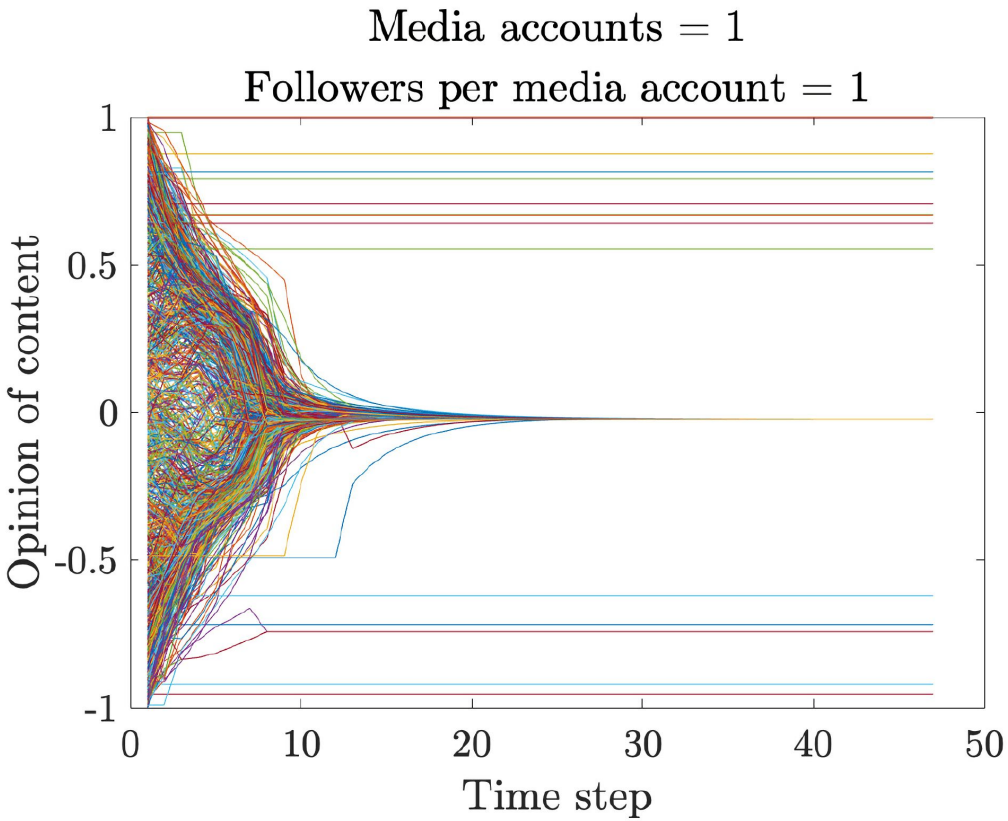




Blue means low impact and red means high impact.

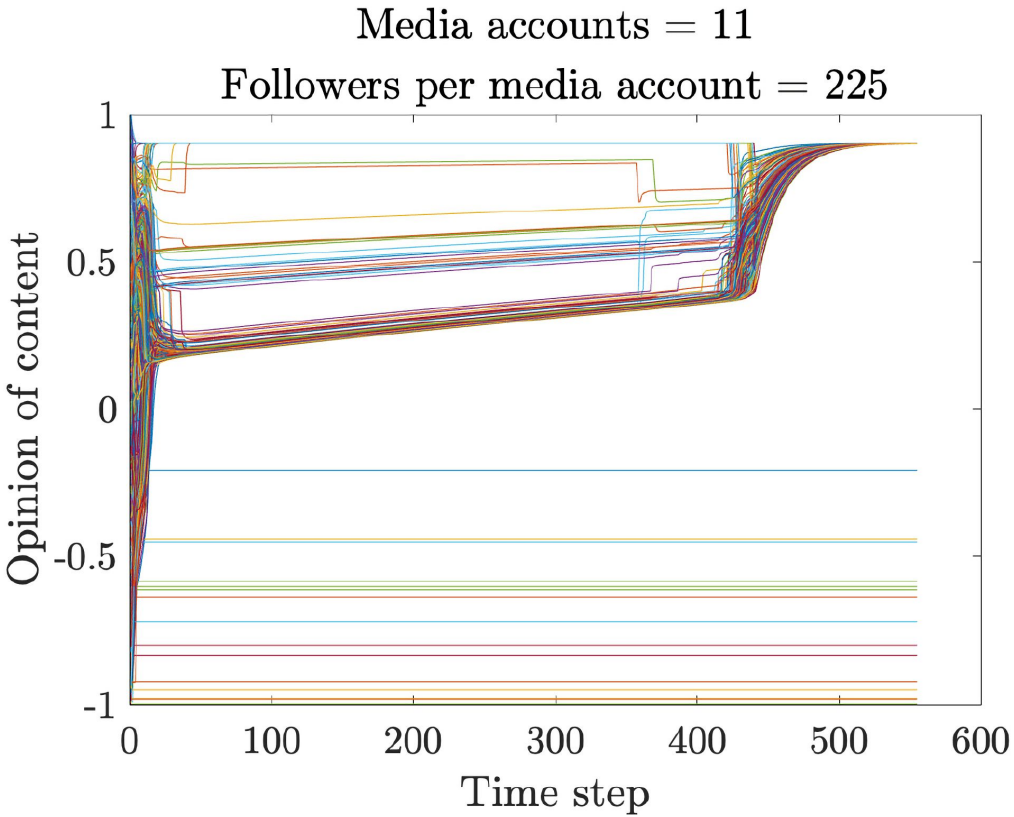
[Brooks, Porter (preprint 2019)]

Evolution of opinions - few media, few followers



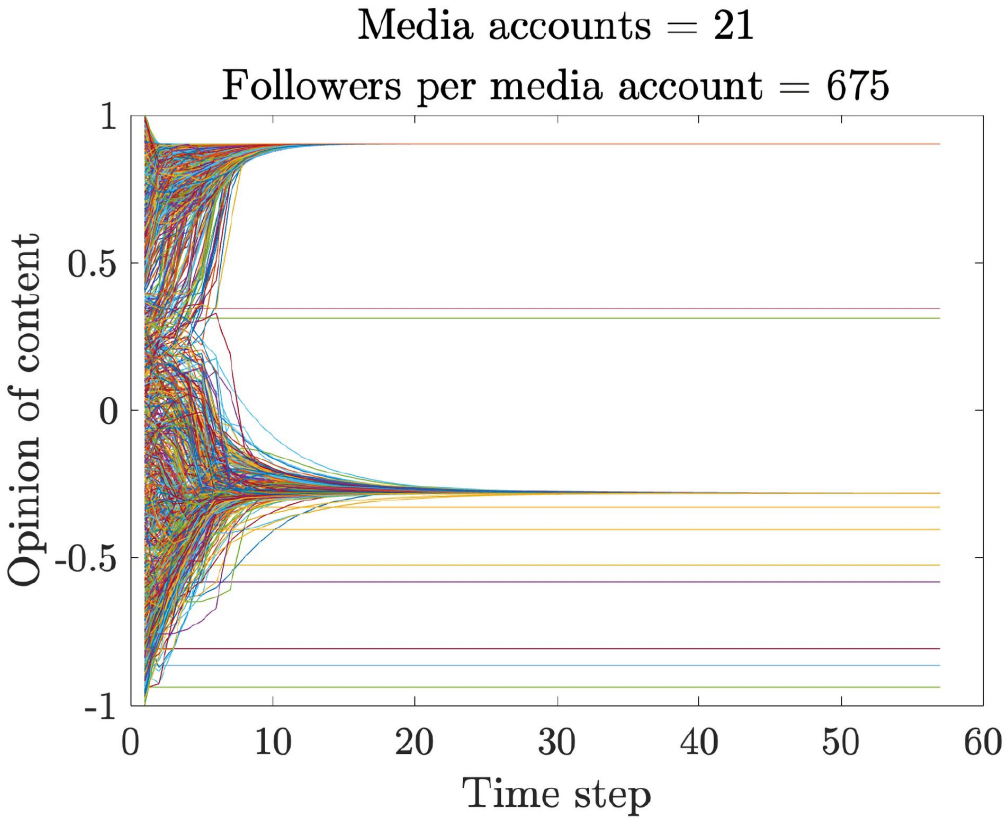
[Brooks, Porter (preprint 2019)]

Evolution of opinions - some media, some followers



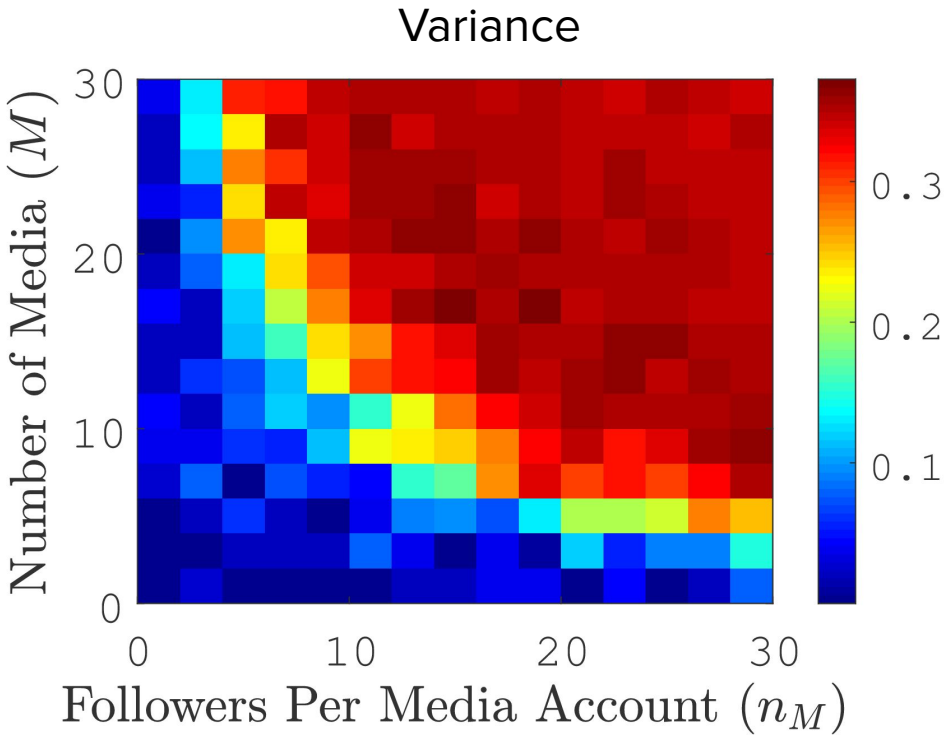
[Brooks, Porter (preprint 2019)]

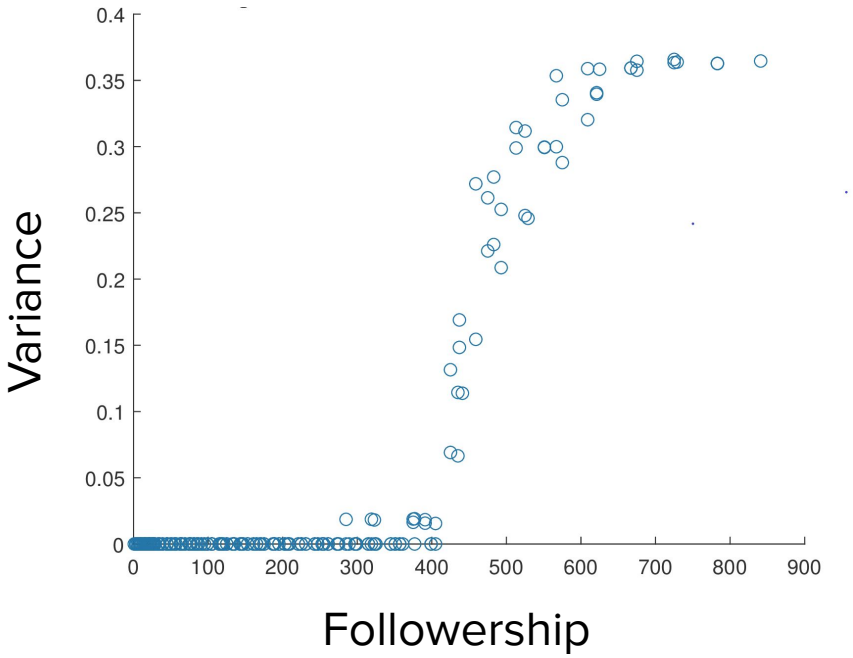
Evolution of opinions - high media, high followers



Measuring media impact with variance

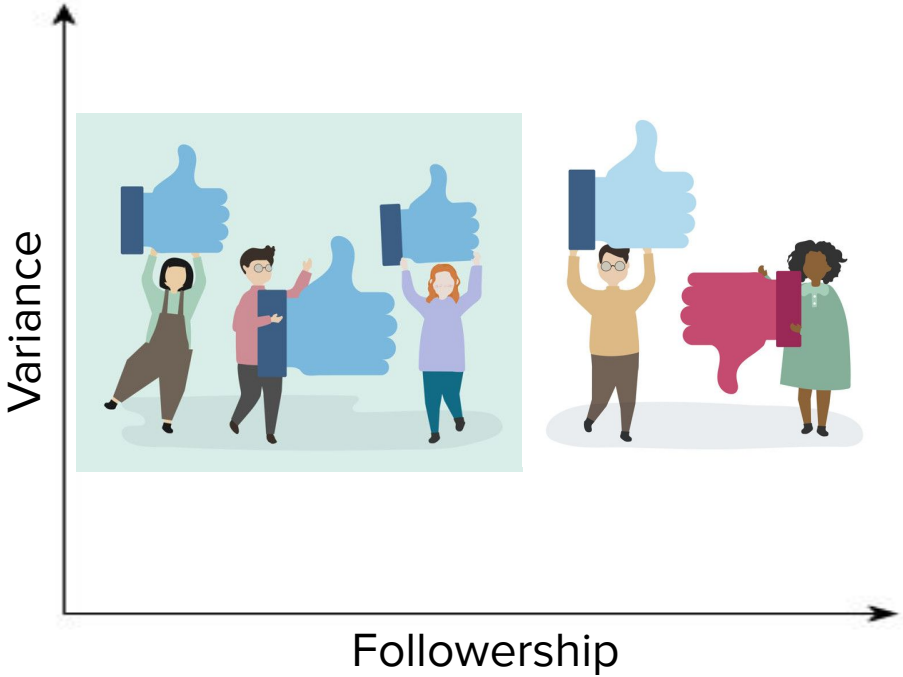
How do we measure how polarized the population is?

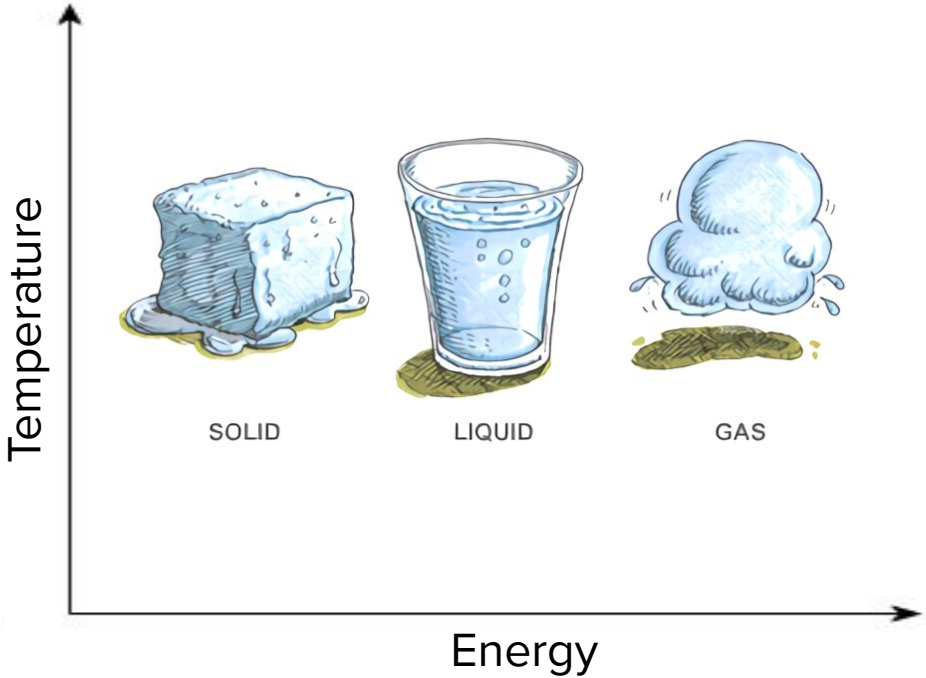




[American Water Works Association (2002)]
[Freepik]

Bifurcations





Bifurcation analysis

$$x_i^k \longrightarrow x_i(t) \longrightarrow \rho(t,x)$$

Discrete-in-time	Continuous-in-time	Continuous-in-time
Discrete-in-space	Discrete-in-space	Continuous-in-space

$$\frac{x_i^{\tau,k+1} - x_i^{k,\tau}}{\tau} = -\frac{\frac{1}{N}\sum_{j=1}^{N}\left(x_i^{\tau,k} - x_j^{\tau,k}\right)\eta\left(\left|x_i^{\tau,k} - x_j^{\tau,k}\right|\right) + \frac{\alpha_i}{N}\left(x_i^{\tau,k} - m\right)\eta\left(\left|x_i^{\tau,k} - m\right|\right)}{\frac{1}{N}\sum_{j=1}^{N}\eta\left(\left|x_i^{\tau,k} - x_j^{\tau,k}\right|\right) + \frac{\alpha_i}{N}\eta\left(\left|x_i^{\tau,k} - m\right|\right)}$$

$$\partial_t \rho = \operatorname{div}\left(\rho\left(\frac{\int \eta\left(|x-y|\right)(x-y)\rho(t,y)\,dy + (x-m)\eta\left(|x-m|\right)b(x)}{\int \eta\left(|x-y|\right)\rho(t,y)\,dy + \eta\left(|x-m|\right)b(x)}\right)\right)$$

Summary of SURF work

Investigated...

- Several measures for media impact
- Different network structures
- Different bounded-confidence models
- Different initial conditions to the model (media and individuals)
- Different methods of increasing media followership
- Continuum limit of model and derived macroscopic model

Future work

- Rigorous proof of the continuum model
- Stationary states of the continuum model
- Variations to the model:
  - Extend to higher dimensions; allows for more nuanced opinions.
  - Have multiple media ideologies; draw the media from an empirical distribution.
  - Make the confidence interval a function of the opinion.
- Explore other measures of media impact.

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